

SECTION A

1 (**short**) An inextensible flexible cable is suspended between two rigid arms as shown in Fig. 1. Both arms can pivot freely and form angle θ to the horizontal. The cable is loaded as shown by a continuously distributed uniform load of ω per unit horizontal distance.

(a) Show that the force in each arm has magnitude $R = \omega L / \sin \theta$. [4]

(b) Derive an expression for the shape $v(x)$ of the cable in terms of x , θ and L using the co-ordinate system shown in the figure. [6]

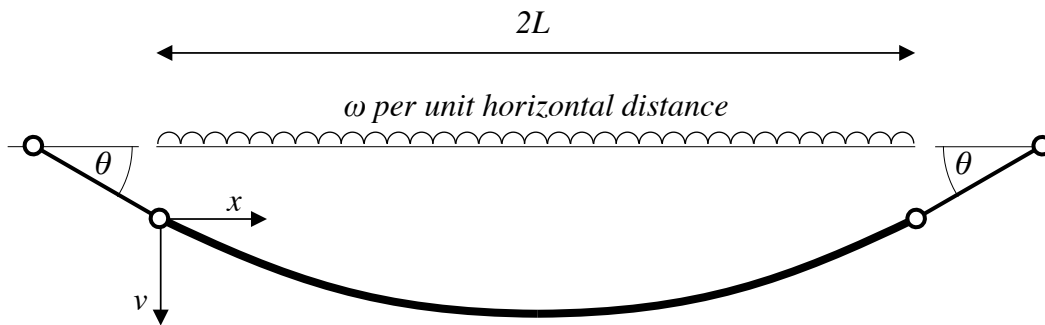


Fig. 1